User:

"These are the recent opinions from other agents: \n"

"One of the agents' response:\n"

{agent 2's response from the previous round}

"One of the agents' response:\n"

{agent 3's response from the previous round}

"This was your most recent opinion: \n"

{agent 1's response from the previous round}

"Use these opinions carefully as additional advice to revise your recent opinion to give your final answer to the question:\n"

{Question}

Figure 3. Prompt for Multi-Agent evaluation. Three agents debate is shown for brevity.

D. Additional Experiments

D.1. Training Dynamics of SDRL

To better understand SDRL training dynamics, we visualize (i) the number of constructed debate prompts and (ii) the number of debate-conditioned responses used at each prompt-generation step on the training set. The results are shown in Figure 4. In the figure, *Filter* refers to our filtering procedure that removes debate responses with zero advantage. *Debate Prompts* denotes the number of debate prompts retained for policy updates, and *Debate Samples* denotes the number of debate-conditioned responses used for policy updates. The results indicate that most debate training occurs early in training. As training progresses, debating on the training set becomes easier, and an increasing fraction of debate responses receives zero advantage, providing no learning signal. Since we do not apply oversampling for debate responses, the additional computational cost of SDRL remains modest.

In the bottom row, *Debate Accuracy* denotes the accuracy of second-turn responses conditioned on debate contexts, and *Debate Accuracy Improvement* measures the accuracy gain of second-turn responses over the initial responses. These trends suggest that SDRL effectively equips the model with debate capability, enabling it to revise answers productively when exposed to alternative reasoning trajectories.

D.2. Reduced Training Budget

As SDRL requires additional computation to train debate pairs, we also evaluate SDRL under a reduced training budget on Qwen3-4B-Base. Table 5 reports results for SDRL+freq trained with only 125 prompt-generation steps, which uses fewer training resources than the DAPO baseline trained with 200 steps. Despite the reduced budget, SDRL+freq still outperforms the well-trained DAPO baseline in debate performance, while remaining worse than the fully trained SDRL, indicating that SDRL has not saturated at 125 steps. These results further support that the gains of SDRL come from training on debate data, rather than simply from increased overall training compute.

D.3. Case Study

We present a case study illustrating performance degradation on MATH500 over multiple debate rounds using the Qwen3-4B-Base model trained with SDRL. For this example, the correct answer is $2k + 2$. Figure 5, Figure 7, and Figure 6 show the detailed responses of the five agents under the decentralized MAD setting for the initial responses, debate round 1, and debate round 2, respectively. As shown, responses that yield the correct answer are substantially shorter than those that produce an incorrect final answer. After debate, more agents shift toward the incorrect answer accompanied by longer reasoning traces.

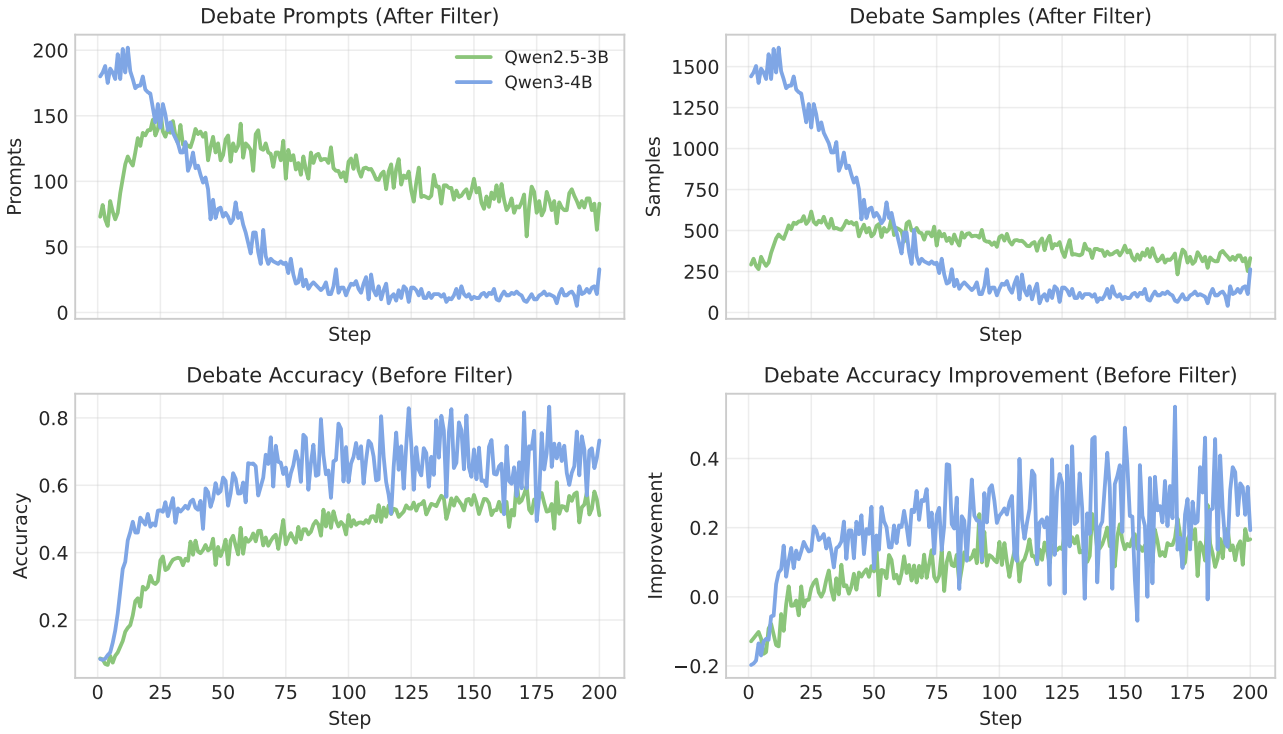

Figure 4. Training dynamics of debate-related values using SDRL.

Table 5. Reduced training budgets for SDRL+freq with different numbers of training steps, evaluated in the decentralized multi-agent debate setting. The debate system contains 5 agents. *Maj* denotes the majority-vote accuracy of the agents' direct responses to the question. *Debate* denotes the performance of the decentralized multi-agent system after debate round 1. Δ is the difference between *Maj* and *Debate*. The best results in each column are bolded.

Method	MATH500			AMC23			AIME24			AIME25			Avg.		
	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ	Maj	Debate	Δ
Qwen3-4B-Base															
DAPO-200 steps	86.9	83.1	-3.8	**73.0**	76.0	3.0	26.7	28.3	1.6	24.7	24.0	-0.7	52.8	52.9	0.1
+ SDRL-125 steps	86.6	**86.0**	**-0.6**	68.5	78.0	**9.5**	24.7	32.0	**7.3**	27.3	27.3	0.0	51.8	55.8	**4.0**
+ SDRL-200 steps	**87.0**	85.9	-1.1	72.5	**79.0**	6.5	**28.7**	**36.0**	**7.3**	**28.7**	**30.0**	**1.3**	**54.2**	**57.7**	3.5

We hypothesize that this failure mode arises because MATH500 is relatively easy for Qwen3-4B, and the model often retrieves the correct answer primarily from memorized knowledge acquired during pre-training. In this case, long but incorrect reasoning traces can dominate the shared debate context, biasing subsequent generations toward the same incorrect trajectory. To support this observation, Table 6 reports the average response lengths for correct and incorrect answers across debate rounds. The results show that incorrect responses are consistently longer than correct responses, which correlates with the observed accuracy drop as debate proceeds.

E. Additional Proofs for Section 4

This appendix provides detailed proofs for the results in Section 4. Our analysis follows the DCM-based treatment in Choi et al. (2025), and extends it by incorporating a private critique pseudo-count vector β (Definition 4.1).

E.1. Notation and standing assumptions

Fix an input question q and a finite answer set $\mathcal{A} = \{1, \ldots, K\}$. Without loss of generality, index 1 is the correct answer. For each agent i and round t, let $\alpha_{i,t} \in \mathbb{R}_+^K$ be Dirichlet pseudo-counts and define the Dirichlet mean $\bar{\theta}_{i,t} := \alpha_{i,t}/\|\alpha_{i,t}\|_1 \in \Delta^K$.

Agent1: **Identify the terms in the expression:**\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n\n2. **Distribute the addition operation:**\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n This can be rewritten as:\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n3. **Combine like terms:**\n - Combine the \\(k \\) terms: \\(-k + 3k \\)\n - Combine the constant terms: \\(4 - 2 \\)\n\n Let's do these steps separately.\n\n4. **Combine the \\(k \\) terms:**\n \\[\n -k + 3k = 2k\n \\]\n\n5. **Combine the constant terms:**\n \\[\n 4 - 2 = 2\n \\]\n\n6. **Combine the results:**\n \\[\n 2k + 2\n \\]\n\nSo, the simplified expression is:\n\\[\n2k + 2\n\\]\n\nLet's confirm this by using Python code.\n```python\nfrom sympy import symbols, simplify\r\n\r\n# Define the variable\r\nk = symbols('k')\r\n\r\n# Define the expression\r\nexpression = (-k + 4) + (-2 + 3*k)\r\n\r\n# Simplify the expression\r\nsimplified_expression = simplify(expression)\r\nprint(simplified_expression)\n```\n```output\n2*k + 2\n```\nThe simplified expression is indeed \\(2k + 2 \\).\n\nTherefore, the final answer is:\n\\[\n\\boxed{2k + 2}\n\\]",

Agent2: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), follow these steps:\n\n### Step 1: Remove the Parentheses\nFirst, remove the parentheses by distributing the addition sign:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\n### Step 2: Combine Like Terms\nNext, combine the like terms. Like terms are those that contain the same variable or are constants.\n\n- Combine the \\(k\\) terms:\n \n \\[\n -k + 3k = 2k\n \\]\n\n- Combine the constant terms:\n \n \\[\n 4 - 2 = 2\n \\]\n\n### Step 3: Write the Simplified Expression\nNow, put the combined terms together:\n\n\\[\n2k + 2\n\\]\n\n### Final Answer\nThus, the simplified form of the expression is:\n\n\\[\n\\boxed{2k + 2}\n\\]",

Agent3: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), follow these steps:\n\n### Step 1: Remove the Parentheses\nFirst, remove the parentheses by distributing the positive sign (since both terms inside the parentheses are already positive):\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\n### Step 2: Combine Like Terms\nNext, combine the like terms. Like terms are those that contain the same variable or are constants.\n\n- Combine the \\(k\\) terms:\n \\[\n -k + 3k = 2k\n \\]\n \n- Combine the constant terms:\n \\[\n 4 - 2 = 2\n \\]\n\n### Step 3: Write the Simplified Expression\nPutting it all together, the simplified expression is:\n\n\\[\n2k + 2\n\\]\n\n### Final Answer\n\\[\n\\boxed{2k + 2}\n\\]",

Agent4: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), follow these steps:\n\n1. **Remove the parentheses**:\n \\[\n (-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n \\]\n\n2. **Combine like terms**:\n - Combine the \\(k\\) terms: \\(-k + 3k = 2k\\)\n - Combine the constant terms: \\(4 - 2 = 2\\)\n\n3. **Write the simplified expression**:\n \\[\n 2k + 2\n \\]\n\nSo, the simplified form of \\((-k + 4) + (-2 + 3k)\\) is:\n\n\\[\n\\boxed{2k + 2}\n\\]",

Agent5: " What is the coefficient of k? To simplify the expression \\((-k + 4) + (-2 + 3k) \\), follow these steps:\n\n1. **Remove the Parentheses:**\n \n \\[\n (-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n \\]\n\n2. **Combine Like Terms:**\n \n - **Combine the \\(k \\)-terms:**\n \n \\[\n -k + 3k = 2k\n \\]\n \n - **Combine the constant terms:**\n \n \\[\n 4 - 2 = 2\n \\]\n \n3. **Write the Simplified Expression:**\n \n \\[\n 2k + 2\n \\]\n\n4. **Identify the Coefficient of \\(k \\):**\n \n The coefficient of \\(k \\) is the number multiplied by \\(k \\), which is **2**.\n\nTherefore, the coefficient of \\(k \\) is:\n\n\\[\n\\boxed{2}\n\\]"

Figure 5. Initial responses on MATH500 with 5 agents.

Table 6. Average response length (tokens) of correct vs. incorrect answers across debate rounds for the MATH500 case study.

Round	Correct length	Incorrect length
Initial (Round 0)	2227	7121
Debate Round 1	2582	6189
Debate Round 2	3027	6200
Debate Round 3	3256	5951
Debate Round 4	3412	5826

The belief in the correct answer is

$$p_{i,t} := \bar{\theta}_{i,t}^{(1)} = \frac{\alpha_{i,t}^{(1)}}{\|\alpha_{i,t}\|_1}.$$

DCM marginal. Under Definition 4.2, the DCM sampling scheme implies the following marginal: for any $k \in \{1, \ldots, K\}$,

$$\Pr(y_{i,t} = k \mid \alpha_{i,t}) = \bar{\theta}_{i,t}^{(k)} = \frac{\alpha_{i,t}^{(k)}}{\|\alpha_{i,t}\|_1}. \tag{13}$$

Neighbor counts. Let $N(i)$ denote the neighbor set of agent i at round t (including itself, if desired). Given neighbors' previous responses $\{y_{j,t-1}\}_{j \in N(i)}$, define the count vector $c_{i,t} \in \mathbb{N}^K$ by

$$c_{i,t}^{(k)} := \sum_{j \in N(i)} \mathbb{1}\{y_{j,t-1} = k\}. \tag{14}$$

Because $c_{i,t}$ counts exactly $|N(i)|$ responses,

$$\|c_{i,t}\|_1 = \sum_{k=1}^{K} c_{i,t}^{(k)} = |N(i)|. \tag{15}$$